

Global Macro Regime Detection & FICC Allocation: A Fully Data-Driven Bayesian Framework

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Abstract—The paper provides a purely data-driven framework for macro-regime detection and asset allocation across Fixed Income, Currencies, and Commodities (FICC). The time dynamics component is provided by a Hidden Markov Model (HMM) with equity features on a global scale, which uses its transition matrix to capture time dynamics. Inference under Bayesian theory with respect to an 11-dimensional vector of macro evidence (CPI, NFP, 10-year bond yield) allows calculating the posterior regime probabilities. Real economy based regimes make up a 2-by-2 classification via growth and inflation. All parameters in the framework are estimated using data, with no parameter assigned ad hoc. Asset weights are determined by regime-conditional Sharpe ratios or information ratios computed against an equal-weight FICC benchmark, validated through a walk-forward backtest with a strict no-look-ahead protocol.

Index Terms—Macro regime detection, Hidden Markov Model, Bayesian inference, FICC allocation, CPI, Non-Farm Payrolls, information ratio, walk-forward backtest

I. INTRODUCTION

FICC returns are heavily impacted by the state of the economy. Government bond prices rise when central banks lower interest rates in times of deflationary recessions. Gold tends to outperform stocks amid stagflation conditions because investors search for inflation hedges while growth declines. Credit spreads tighten when there is steady growth and low inflation in Goldilocks regime. Determining the type of regime that exists and then deploying capital accordingly is key for multi-asset investments.

There is a large amount of research regarding regime identification. Hamilton’s pioneering model of Markov switching [1] showed that economic variables experience changes in their dynamics. This research provided the foundation for regime-informed portfolio allocations [2], [3]. The theory was later extended with more complex state spaces, time-varying transition probabilities and Bayesian inference [4]. Nonetheless, most applications used some kind of hand-coding of parameters, such as by equating stock volatility with inflation, applying certain weights between trend slope and growth rate, or setting up payoff matrices for asset allocation based on economic sense rather than data.

This paper proposes a fully data-driven framework for macro regime detection and FICC allocation that eliminates all hand-coded parameters. The framework consists of three components, each estimated entirely from data:

- 1) **Regime definition.** Instead of employing proxy variables, such as equity volatility to represent inflation, the growth–

inflation regime classification is based on economic data directly: CPI YoY for the inflation axis and NFP trend for the growth axis. The expanding median cutoff rule (Section III) guarantees that no future information is used in the classification procedure.

- 2) **Regime inference.** This paper uses a Gaussian HMM trained on global equity features to capture regime transition dynamics, providing a data-driven prior via its transition matrix. This prior is updated daily, using an 11-dimensional macro evidence vector as the likelihood, through Bayes’ theorem.
- 3) **Portfolio allocation.** Rather than specifying a required asset performance via some payoff matrix, weights for each regime are determined by regime-conditional information ratios [5] computed against an equal-weight FICC benchmark. For comparison, a Sharpe-ratio-based variant is also reported. Weights are estimated at each rebalancing period using only information available prior to the rebalance. The approach is backtested using a walk-forward backtest with strict no look-ahead constraints over the time period 2018–2026. In particular, this encompasses the pre-COVID expansion, the 2020 recession, the 2021–2022 inflationary episode, and the tightening cycle afterwards.

II. DATA SOURCES

All data is sourced from two open-source providers: Yahoo Finance via the `yfinance` Python package, and the Federal Reserve Economic Data (FRED) API. Figure 1 shows the end-to-end system architecture built on these data sources.

A. Yahoo Finance

a) *Global Equity Indices.*: Daily closing prices for six equity indices are collected from 2018 to the present (Table I), covering the US, UK, Germany, France, Japan, and Austria.

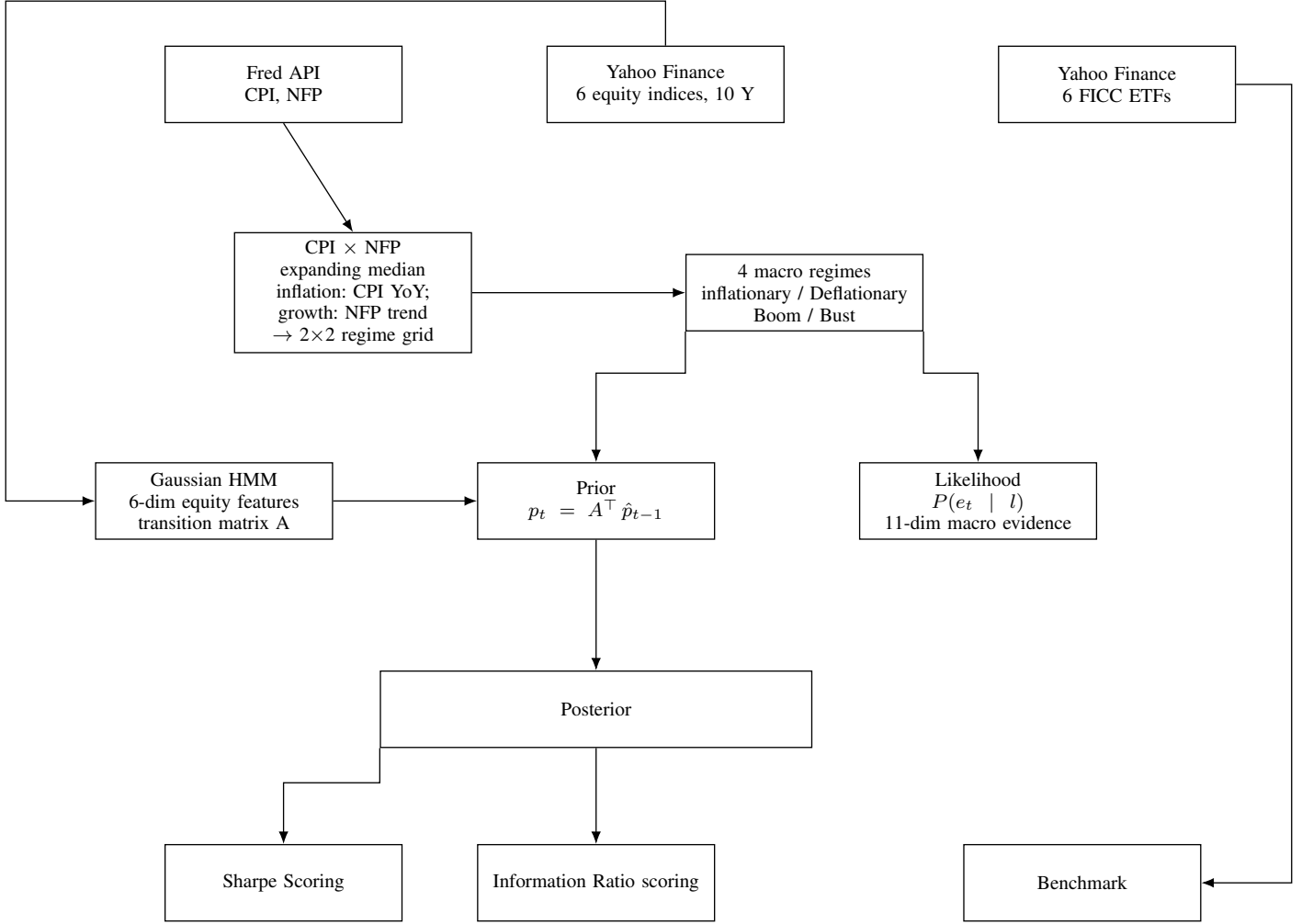


Fig. 1: System architecture. Data flows top-to-bottom through four stages: (0) raw data acquisition, (1) walk-forward regime labeling from CPI × NFP expanding medians, (2) Bayesian regime inference combining an HMM-derived prior with an 11-dimensional macro-evidence likelihood, and (3) regime conditional allocation under two scoring rules (Sharpe and Information Ratio), both feeding (4) a monthly rebalancing walk-forward backtest against an equal-weight benchmark. FICC ETF prices bypass the inference stages and enter the backtest directly.

TABLE I: Equity indices used for HMM feature construction.

Market	Index	Ticker
United States	S&P 500	GSPC
United Kingdom	FTSE 100	FTSE
Germany	DAX	GDAXI
France	CAC 40	FCHI
Japan	Nikkei 225	N225
Austria	ATX	ATX

TABLE II: FICC instruments used in the backtest.

Asset	ETF	Tracks
Gov Bonds	TLT	US 20+ Year Treasury
US Dollar	UUP	DB USD Index
Gold	GLD	Gold spot
Crude Oil	USO	WTI crude
HY Credit	HYG	iBoxx HY Corporate
Cash	BIL	1–3 Month T-Bill

c) *10-Year Treasury Yield.*: The US 10-year Treasury yield ($\hat{TNX}$) is retrieved at daily frequency and used as part of the macro evidence vector.

b) *FICC Instrument Prices.*: Daily closing prices for six ETFs covering FICC asset classes are used in the backtest (Table II).

B. FRED (Federal Reserve Economic Data)

Two monthly time series are downloaded via the FRED API:

- **CPI** (CPIAUCSL). Consumer Price Index for all urban consumers, seasonally adjusted. Year-over-year percentage change is used on the inflation axis for regime labeling.
- **NFP** (PAYEMS). Total Non-Farm Payroll Employment, seasonally adjusted. The 3-month moving average of month-over-month changes (NFP trend) is used on the growth axis.

Both monthly series are forward-filled to daily frequency. Data collection begins in 2015 to provide sufficient history for the expanding medians, while backtesting starts from 2018.

III. MACRO REGIME DEFINITIONS

A. The 2-by-2 Growth–Inflation Framework

The majority of macroeconomic cycle models stem from the assumption that asset behavior depends not on growth or inflation in isolation but on the combination relative to their own historical trends. For instance, a bond rally during a recession differs fundamentally from a bond sell-off in stagflation, despite both being associated with weak activity. This motivates a 2×2 classification (Table III).

TABLE III: The four macro regimes and their FICC implications.

Regime	Growth	Infl.	Asset implications
Inflationary Boom	↑	↑	Oil, credit bid; bonds hurt
Deflationary Boom	↑	↓	Credit strongest; cash worst
Inflationary Bust	↓	↑	Gold, USD; credit crushed
Deflationary Bust	↓	↓	Bonds rally; oil collapses

The four cells are not equally frequent, since the economy spends more time in stable regimes (inflationary boom, deflationary boom, deflationary bust) than transitional regimes (inflationary bust). But the transitions matter most for portfolio construction, as missing a shift from deflationary boom to inflationary bust (which happened in 2021–2022) can be catastrophic for a fixed allocation strategy.

B. Why CPI and NFP?

An earlier version of this framework used rolling equity volatility as a stand-in for inflation and the Kalman trend slope as a measure of growth. Both are market-derived and are available at a daily frequency, making them seemingly effective options. But the proxy relationship breaks down when it matters most: Equity volatility signals uncertainty, not inflation. During the 2020 COVID crash, volatility spiked to levels last seen in 2008 while CPI was falling. A volatility-based proxy would have labelled that episode as stagflation. The Kalman slope has a similar issue as

it captures price momentum, not economic output. For example, steep upward slope may simply reflect a post-recession relief rally.

CPI and NFP avoid both of the issues stated above. CPI measures the price level directly, while NFP measures labour market activity—the most reliable near-term proxy for aggregate demand. The monthly release frequency for these data is acceptable because macroeconomic regimes themselves evolve on monthly timescales.

C. Expanding Median and the No-Look-Ahead Constraint

A regime threshold needs a reference level. A fixed constant (e.g., $\text{CPI} > 3\%$ being some regime) imposes a bias that may be inappropriate across as time goes on. A full-sample median, on the other hand, leaks future data into past classifications, making backtests unreliable.

The expanding median resolves both problems. For each date t , let

$$m_t = \text{Median}(x_1, x_2, \dots, x_t). \quad (1)$$

Using Table III, we may then specify a regime label r_t into one of the four with the expanding medians for CPI and NFP serving as thresholds for inflation and growth, respectively. Since m_t depends only on $\{x_s\}_{s \leq t}$, no future information enters. The threshold stabilises as data accumulates, and the first 60 months are discarded to avoid noisy estimates during cold-start. The median is recomputed at every backtest rebalance date, reproducing what a manager can observe in real time.

IV. MODEL

The inference pipeline has four stages: (i) Kalman filtering extracts trend signals from equity prices; (ii) a Gaussian HMM learns regime transition dynamics; (iii) a macro evidence vector captures the current economic state; (iv) Bayesian updating produces the daily posterior regime distribution.

A. Kalman Filter — Trend Extraction

For each equity index i , a Local Linear Trend model is fitted to log-prices $y_{i,t} = \log P_{i,t}$:

$$y_{i,t} = \mu_{i,t} + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma_\varepsilon^2), \quad (2)$$

$$\mu_{i,t} = \mu_{i,t-1} + \nu_{i,t-1} + \xi_t, \quad \xi_t \sim \mathcal{N}(0, \sigma_\xi^2), \quad (3)$$

$$\nu_{i,t} = \nu_{i,t-1} + \zeta_t, \quad \zeta_t \sim \mathcal{N}(0, \sigma_\zeta^2), \quad (4)$$

where $\mu_{i,t}$ is the smoothed level and $\nu_{i,t}$ the trend slope. Cross-sectional median slope $\tilde{\nu}_t$ and dispersion $\sigma_{\nu,t}$ are passed to the HMM.

B. Gaussian HMM — Transition Dynamics

A four-state Gaussian HMM is fitted to:

$$\mathbf{x}_t = [\tilde{u}_t, \Sigma_{v,t}, \tilde{\nu}_t, \sigma_{\nu,t}, \tilde{r}_t, \tilde{r}_t^{(20)}]^\top \in \mathbb{R}^6,$$

where $\tilde{u}_t, \Sigma_{v,t}$ are the cross-sectional median and dispersion of 60-day rolling volatility; $\tilde{r}_t$ is the cross-sectional

median return; and $\bar{r}_t^{(20)}$ is 20-day smoothed momentum. Parameters are estimated via Baum–Welch EM initialised with k -means clustering (max 200 iterations, $\epsilon = 10^{-4}$). Each HMM state k is matched to the CPI×NFP regime with maximum temporal overlap:

$$l^*(k) = \arg \max_{l \in \mathcal{L}} \sum_t [\hat{z}_t = k] \cdot [r_t = l],$$

where $\mathcal{L}$ is the set of possible macro regimes, and $[\hat{z}_t = k] \cdot [r_t = l]$ is a product of indicator functions that is 1 if, at time t , the estimated HMM state is k and the regime label is l , and zero otherwise. The HMM's sole contribution to downstream inference is its learned transition matrix $\mathbf{A}$ whose entries are $A_{jk} = P(z_{t+1} = k \mid z_t = j)$, with z_t denoting the HMM's hidden state at time t .

C. 11-Dimensional Macro Evidence Vector

The evidence vector $\mathbf{e}_t \in \mathbb{R}^{11}$ aggregates three sources (Table IV).

TABLE IV: Components of the macro evidence vector $\mathbf{e}_t$.

Block	Feature	Definition
CPI (4)	cpi_yoy	$\frac{\text{CPI}_t - \text{CPI}_{t-12}}{\text{CPI}_{t-12}} \times 100$
	cpi_mom	$\frac{\text{CPI}_t - \text{CPI}_{t-1}}{\text{CPI}_{t-1}} \times 100$
	cpi_surp.	$\text{cpi_mom}_t - \overline{\text{cpi_mom}_{t-12:t}}$
	cpi_accel	$\text{cpi_yoy}_t - \text{cpi_yoy}_{t-1}$
NFP (3)	nfp_chg	$\text{NFP}_t - \text{NFP}_{t-1}$ (thousands)
	nfp_surp.	$\text{nfp_chg}_t - \overline{\text{nfp_chg}_{t-12:t}}$
	nfp_trend	$\frac{1}{3} \sum_{j=0}^2 \text{nfp_chg}_{t-j}$
10Y (4)	yield_lvl	10Y Treasury yield (%)
	yield_1d	$Y_t - Y_{t-1}$
	yield_5d	$Y_t - Y_{t-5}$
	yield_vol	$\text{std}_{20d}(Y_t - Y_{t-1})$

The CPI and NFP series are forward-filled to daily frequency, while the 10Y Treasury yield updates daily between macro releases, providing the evidence vector with a combination of slow-moving macro information and fast-moving market-based rates signal. For each regime l , a multivariate Gaussian is estimated from all historical days in that regime:

$$\hat{\boldsymbol{\mu}}_l = \frac{1}{N_l} \sum_{t: r_t=l} \mathbf{e}_t, \quad (5)$$

$$\hat{\boldsymbol{\Sigma}}_l = \frac{1}{N_l} \sum_{t: r_t=l} (\mathbf{e}_t - \hat{\boldsymbol{\mu}}_l)(\mathbf{e}_t - \hat{\boldsymbol{\mu}}_l)^\top + \lambda \mathbf{I}, \quad (6)$$

with $\lambda = 10^{-4}$ serving as a covariance regularization parameter. The likelihood $P(\mathbf{e}_t \mid l) = \mathcal{N}(\mathbf{e}_t; \hat{\boldsymbol{\mu}}_l, \hat{\boldsymbol{\Sigma}}_l)$ is normalised across regimes.

D. Bayesian Update

Prior. Yesterday's posterior is propagated via the HMM transition matrix to form today's prior

$$\mathbf{p}_t^{\text{prior}} = \mathbf{A}^\top \hat{\mathbf{p}}_{t-1}. \quad (7)$$

Posterior. Prior and likelihood are combined and normalised:

$$\hat{p}_{t,l} = \frac{p_{t,l}^{\text{prior}} \cdot P(\mathbf{e}_t \mid l)}{\sum_{l'} p_{t,l'}^{\text{prior}} \cdot P(\mathbf{e}_t \mid l')}. \quad (8)$$

When prior and likelihood are aligned with one another, the posterior concentrates sharply on one regime; when they do not, there is uncertainty distributed across regimes. Fig. 2 shows a representative output.

V. BACKTEST

In this part, we present a walk-forward backtesting strategy and a data-based allocation approach for FICC. In particular, the regime detection and portfolio construction are carried out using information available up to time τ , where τ is the rebalance date.

A. Protocol

At each month-end τ , the system executes the following steps:

- a) **Regime labelling.** The preceding days until τ are categorized into one of the four regimes based on expanding-median thresholds of CPI YoY and NFP trend. Again, the expanding median guarantees that no data beyond time τ will be used for the categorization process. The regime prevailing at the last trading day of month τ is considered the dominating regime l^* .
- b) **Weight estimation.** For each FICC instrument i , we compute two regime-conditional scoring candidates from all pre- τ days belonging to regime l^* , enabling a head-to-head comparison.

Sharpe score measures each asset's total return per unit of total volatility in isolation:

$$s_i = \frac{\hat{\mu}_i \cdot 252}{\hat{\sigma}_i \cdot \sqrt{252}}, \quad (9)$$

where $\hat{\mu}_i, \hat{\sigma}_i$ are the mean and standard deviation of daily returns in regime l^* .

Information Ratio score [5] follows the Grinold–Kahn active-management framework. Each asset's return is decomposed via OLS against an equal-weight FICC benchmark $r_{B,t}$,

$$r_{i,t} = \alpha_i + \beta_i r_{B,t} + \theta_{i,t}, \quad t \in \{s \leq \tau : r_s = l^*\}, \quad (10)$$

and the regime-conditional information ratio is

$$IR_i^{(l^*)} = \frac{\hat{\alpha}_i \cdot 252}{\hat{\omega}_i \cdot \sqrt{252}}, \quad \hat{\omega}_i = \text{StdDev}(\hat{\theta}_{i,t}), \quad (11)$$

where α_i is the residual return after regressing against the equal-weight benchmark, and ω_i is the residual volatility. Unlike the Sharpe ratio, which treats each asset in isolation, the information ratio isolates the

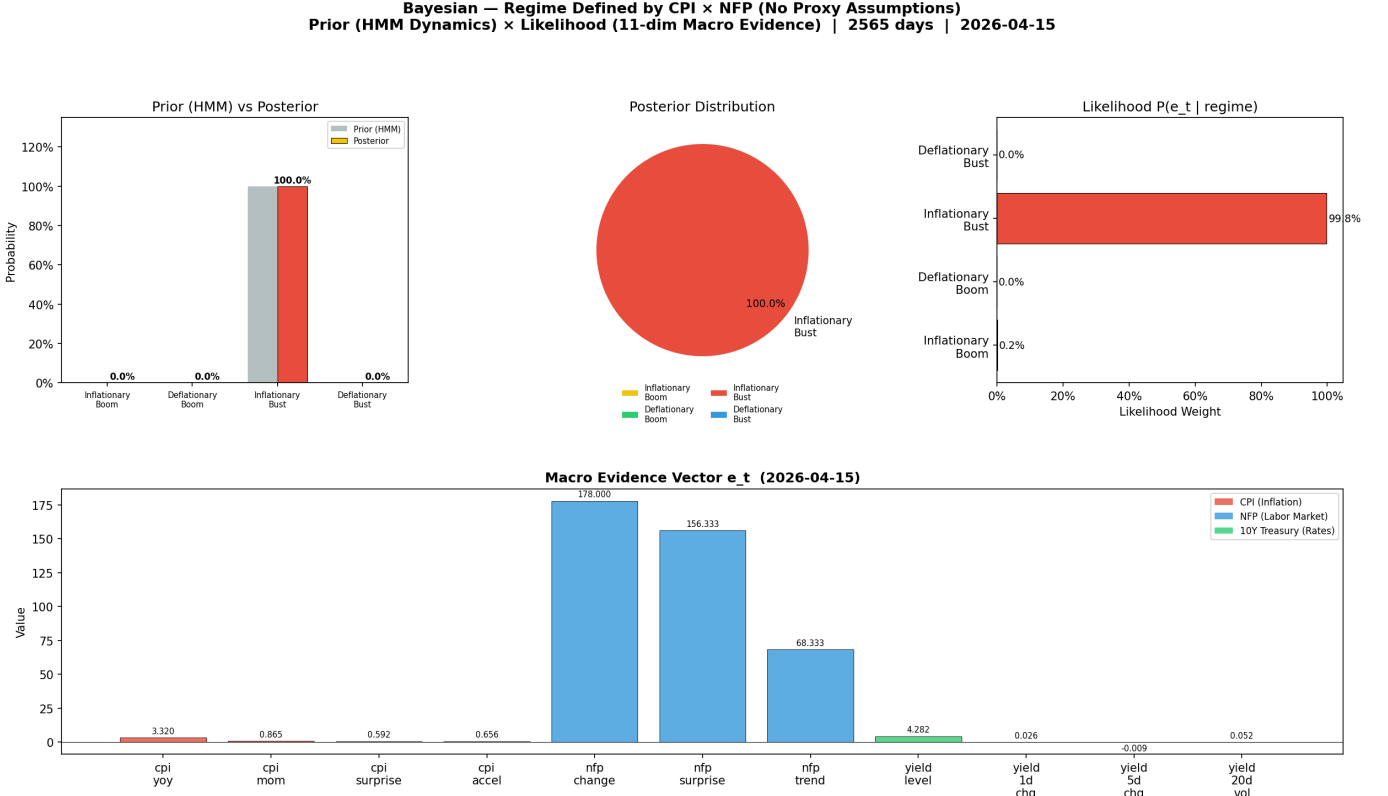


Fig. 2: Bayesian v2 output (2026-04-06). *Top left*: prior (HMM dynamics) vs posterior — Inflationary Bust at 99.4%. *Top centre*: posterior pie. *Top right*: likelihood $P(e_t | l)$ — macro evidence 95.2% consistent with Inflationary Bust. *Bottom*: 11-dim evidence vector colour-coded by block (red = CPI, blue = NFP, green = 10Y).

active contribution of each instrument — its residual return α_i per unit of residual risk ω_i — relative to the uninformed equal-weight allocation.

In both cases, to ensure that the scores are always positive, $|\min_i s_i| + 0.1$ is added. Then, the shifted scores are normalised again to sum to one under the long-only constraint:

$$w_i = \frac{s_i + c}{\sum_j (s_j + c)}, \quad c = |\min_i s_i| + 0.1.$$

- c) **Cold-start fallback.** If regime l^* has fewer than 60 historical trading days at date τ , the portfolio falls back to equal weights across all six instruments.
- d) **Holding period.** The weights generated are stored for one calendar month until the next rebalance cycle.

Notice that three structural protections are implemented against look-ahead leakage:

- a) Expanding-median regime thresholds.
- b) Pre- τ weight estimates using only past returns.
- c) Cold-start default to equal weights if there is not enough regime history.

B. Benchmark

The benchmark is a monthly-rebalanced equal-weight allocation across all six FICC instruments (TLT, UUP, GLD, USO, HYG, BIL), rebalanced on the same month-end dates

as the strategy. This benchmark serves as the reference portfolio B in the information ratio regression of Part V-A.

C. Evaluation Period

The backtest covers the time period January 2018 to April 2026. FICC price data and macro series are collected from 2015 onward to provide three years of history for weight learning prior to the evaluation start date.

VI. RESULTS & DISCUSSION

A. Regime Distribution

Fig. 3 represents the timeline of the macro regime from 2018 to 2026. The first phase, 2018 to 2021, exhibits a quick change between all four regimes, corresponding to the inflationary late-cycle boom and the pandemic shock. Between mid-2021 and late 2022, the model correctly detects a persistent Inflationary Boom due to the surge in demand after the pandemic. Starting from 2023, Inflationary Bust (red) and Deflationary Bust (blue) take turns.

B. Portfolio Weight Learning Mechanism

A key feature of the proposed framework is that portfolio weights are entirely learned from data, rather than a specified beforehand.

Specifically, at each rebalance date τ , only observations $\{t \leq \tau : r_t = l^*\}$ are used, so that for each asset i , we

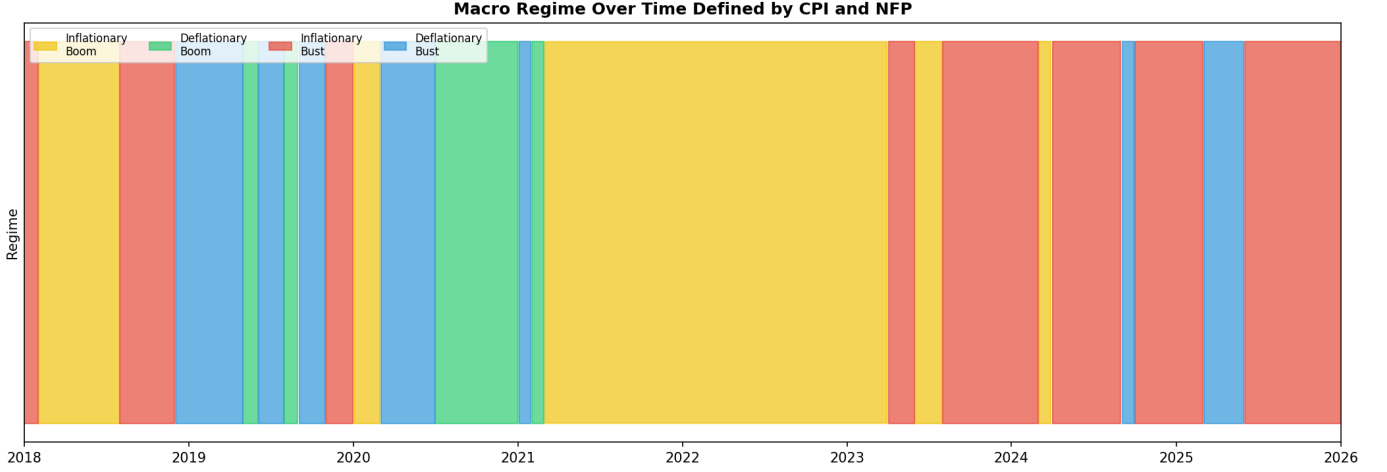


Fig. 3: Macro regime over time defined by $\text{CPI} \times \text{NFP}$ (2018–2026). **Yellow**: Inflationary Boom. **Green**: Deflationary Boom. **Red**: Inflationary Bust. **Blue**: Deflationary Bust.

restrict our attention to historical returns during the same regime l^* , producing a regime-specific dataset:

$$\mathcal{D}_{l^*} = \{r_{i,t} : t \leq \tau, r_t = l^*\}.$$

This reflects the assumption that asset performance is regime-dependent and that past behavior in the same regime is the most relevant indicator for future allocation. Within $\mathcal{D}_{l^*}$, each asset is evaluated using a risk-adjusted metric (Sharpe-based/Information Ratio-based learning given in Section V-A). To ensure a feasible long-only portfolio, scores are transformed into weights w_i via a positivity shift and normalisation. This guarantees $w_i \geq 0$ and $\sum_i w_i = 1$.

If fewer than 60 historical observations are available in regime l^* , the model defaults to equal weights, preventing overfitting in sparse regimes.

This learning mechanism has the interpretation that portfolio allocation updates with new data, such that assets which deliver strong risk-adjusted returns within a regime are overweighted, while those with weaker performance are downweighted. In contrast to static allocation rules, this approach adapts both to changing macro regimes and to evolving cross-asset relationships.

C. Portfolio Weight Dynamics

The weight dynamics (Fig. 4) follow the regime changes closely. Credit (HYG, red) and oil (USO, orange) were dominant investments during the 2018–2020 Inflationary Boom and its initial recovery period. Gold (GLD, yellow) and oil were at their highest in 2021–2022 during the Inflationary Boom. Most importantly, cash (BIL, grey) investments were becoming dominant since 2023. This shift reflects the system’s assessment that the investment environment had deteriorated significantly. This defensive bias led to both underperformance and reduced drawdown risk between 2025–2026.

D. Cumulative Returns

Fig. 6 shows the cumulative returns of three series from 2018 to 2026: the Sharpe-scored strategy (blue), the IR-scored strategy (red), and the equal-weight benchmark (grey dashed). All three track closely until 2022, with both strategies avoiding the benchmark’s -20.4% Q1 2020 draw-down. From 2023 on, the benchmark rallies strongly as bonds, credit, and oil all recover, and its cumulative return reaches 65.5% by April 2026 — about 25 pp above the Sharpe strategy (40.5%) and 29 pp above the IR strategy (36.6%). Both strategies underperform in this credit-led rally because the regime engine persistently labels the environment as Inflationary Bust and hence tilts weights toward cash. The IR strategy is the more conservative of the two, reflecting its sharper penalty on assets whose returns are largely driven by benchmark beta rather than by asset-specific active return.

E. Rolling Sharpe Ratio

Fig. 7 compares 252-day rolling Sharpe ratios for all three series. Through 2022 all three move in similar ranges. The benchmark briefly leads in 2021 on the post-COVID bond rally. Both strategies turn strongly positive from mid-2023, with the IR strategy reaching a rolling Sharpe above 4.8 in early 2025, which is higher than the Sharpe strategy’s peak near 4.4 over the same window. This gap illustrates a consistent pattern throughout the backtest, that the IR-based weights deliver marginally better risk-adjusted performance than Sharpe-based weights, with both decisively outperforming the equal-weight benchmark during the 2023–2026 tightening cycle. Both strategies did dip below zero briefly during the 2021–2022 inflation repricing, with the IR strategy showing slightly deeper short-term drawdowns but recovering faster.

F. Backtest Performance Summary

Full-period statistics are presented in Table V. The two strategies trade off absolute return against risk in subtly

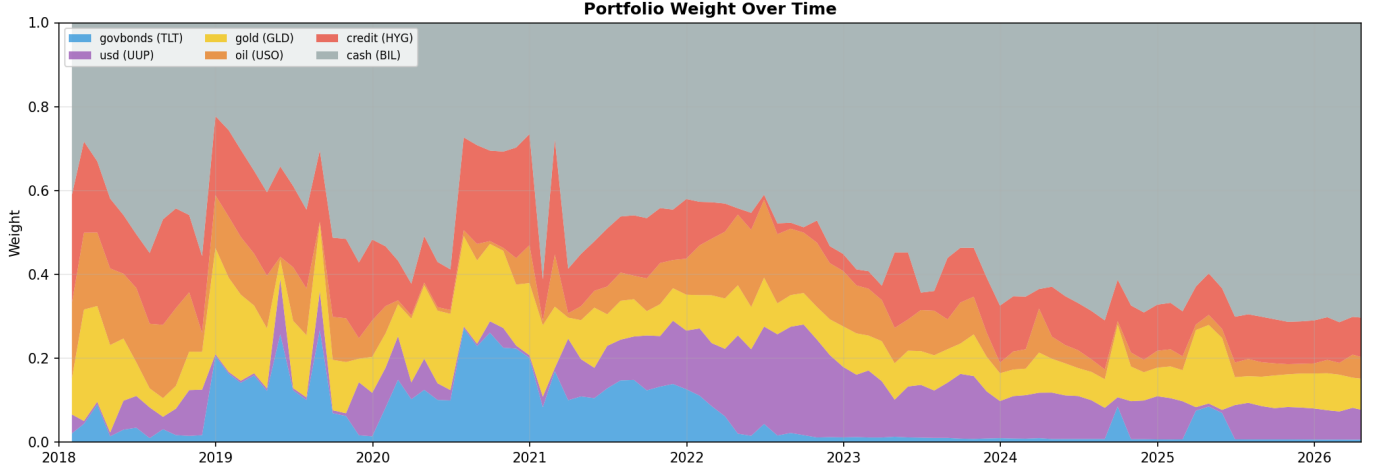


Fig. 4: Portfolio weights over time (learned from data). Cash (BIL, grey) dominates from 2023 as the system detects sustained *Inflationary Bust* and *Deflationary Bust*. Gold and oil peak during 2021–2022 *Inflationary Boom*.

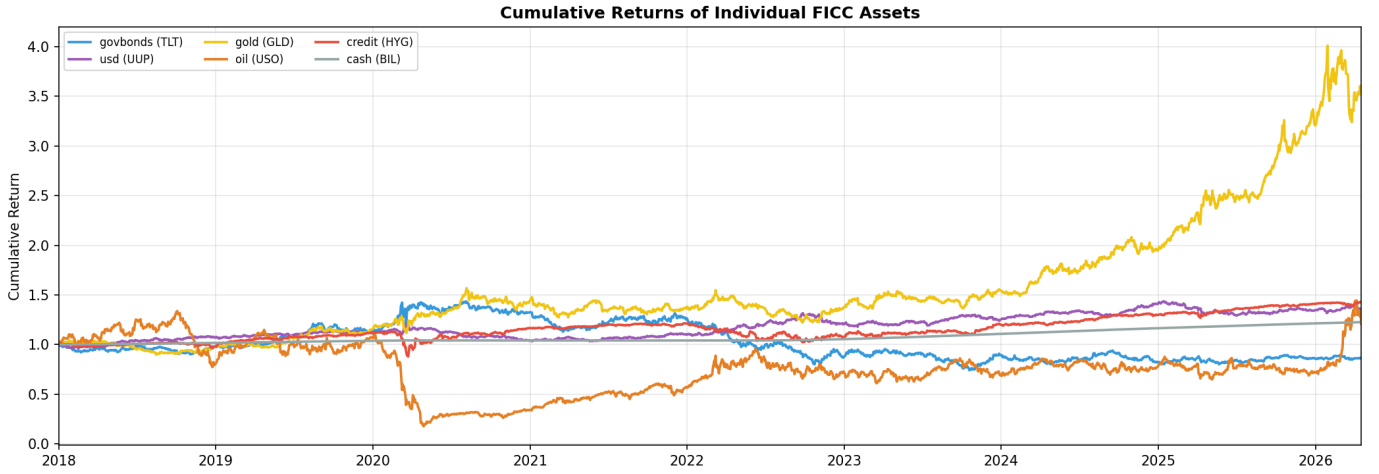


Fig. 5: Cumulative return of each FICC Asset: Gold (GLD, yellow) has outperformed other assets significantly over the time period

different ways, while both outperform the equal-weight benchmark on every risk-adjusted metric.

TABLE V: Backtest performance summary (2018–2026).

Metric	Sharpe	IR	Benchmark
Total Return	40.5%	36.6%	65.5%
Annual Return	4.2%	3.8%	6.3%
Annual Vol	4.4%	3.4%	8.1%
Sharpe Ratio	0.95	1.12	0.78
Max Drawdown	−6.8%	−6.9%	−20.4%
Win Rate	57.4%	57.5%	55.0%

Bold = best among the three series. **Red** = benchmark wins.

Now, the benchmark wins the return race (6.3% annualised vs 4.2% for Sharpe and 3.8% for IR) because the cash-heavy defensive positioning that protects both strategies from drawdowns also excludes them from the 2024–2026 credit rally. However, both strategies dominate on risk-

adjusted measures. The IR strategy achieves the best Sharpe ratio overall (1.12, versus 0.95 for the Sharpe strategy and 0.78 for the benchmark), driven by the lowest realised volatility of the three (3.4%). The Sharpe strategy records the shallowest drawdown (−6.8%), with IR a close second at −6.9%. The −13.6 pp improvement in max drawdown versus the benchmark is the most striking effect of regime-conditional allocation, consistent with its design goal of overweighting the best-performing assets in each regime. Between the two scoring methods, the IR’s higher Sharpe ratio confirms its utility in that by isolating each asset’s active residual return from its benchmark exposure, the information ratio produces a more efficient active allocation than total-return/total-volatility scoring, even though the absolute returns differ by only a few percentage points.

G. Return Attribution

Fig. 8 decomposes each strategy’s total return by instrument. Three observations help compare and contrast Sharpe

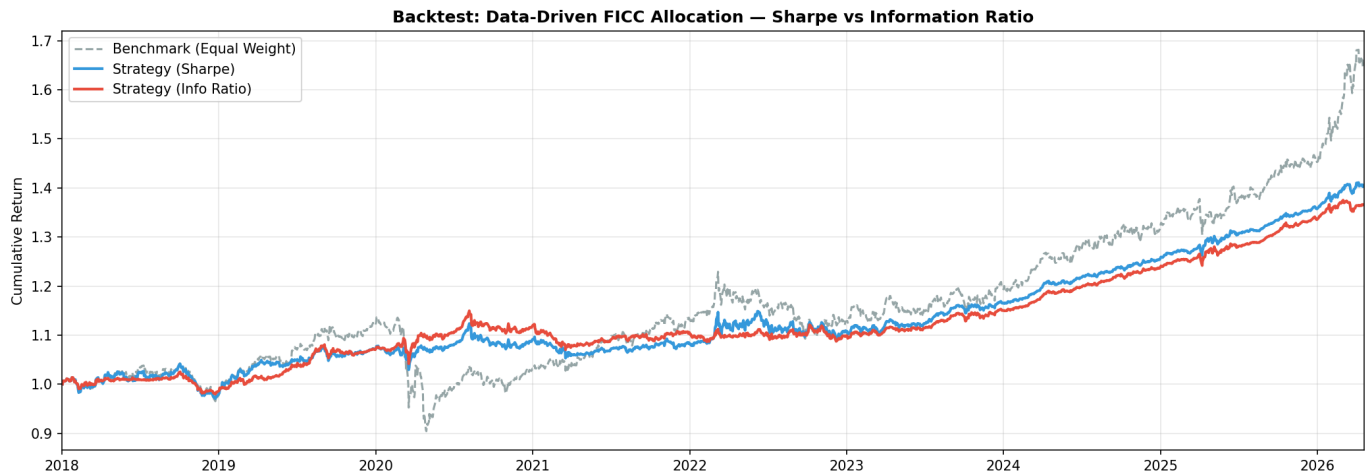


Fig. 6: Cumulative return: Sharpe strategy (blue, solid), IR strategy (red, solid), and equal-weight benchmark (grey, dashed), 2018–2026. Both strategies avoid the 2020 Q1 drawdown but underparticipate in the 2024–2026 credit-led rally.

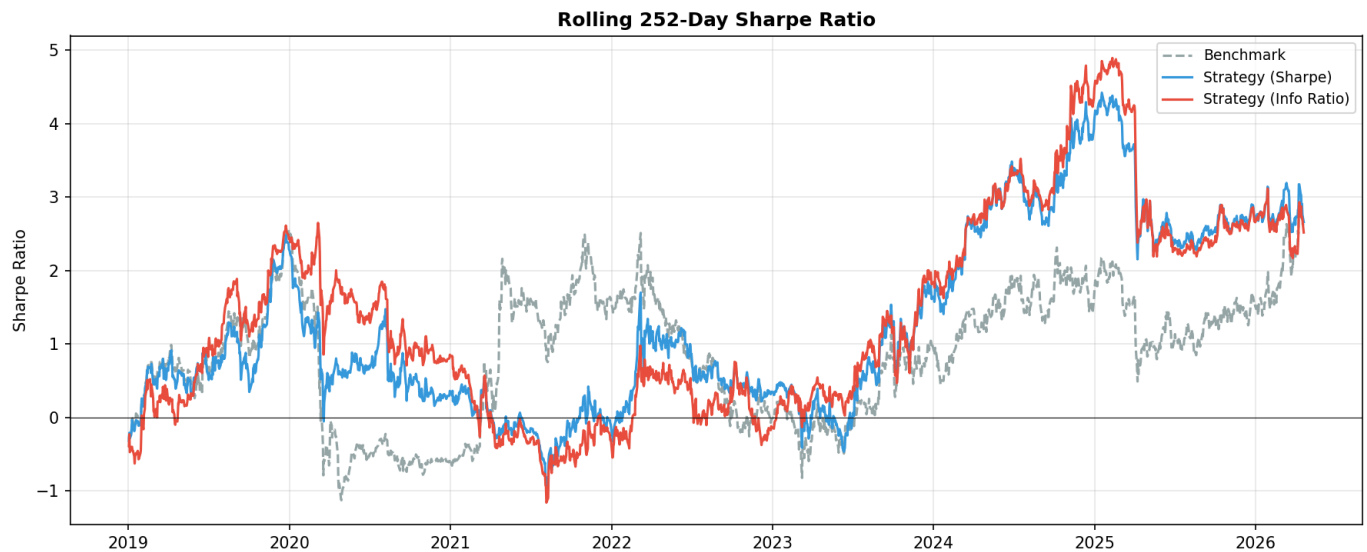


Fig. 7: Rolling 252-day Sharpe ratio: Sharpe strategy (blue, solid), IR strategy (red, solid), and benchmark (grey, dashed). The IR strategy reaches the highest peak (above 4.8 in early 2025), and both strategies lead the benchmark from 2023 onwards.

and IR scoring. Firstly, gold (GLD) is the dominant contributor in both strategies (+11.0% Sharpe, +12.1% IR), reflecting its consistent outperformance in Inflationary Bust periods. Secondly, cash (BIL) contributes roughly +12% in both as the accumulated short-rate yield of the 2023–2026 tightening cycle. Finally, oil (USO) contributes +6.4% under Sharpe but −2.4% under IR. As IR regresses against the benchmark, it recognises that most of oil’s return is explained by its high beta to the equal-weight portfolio rather than by actual residual alpha, and downweights it accordingly. The USD (UUP) position is amplified accordingly (+2.5% under Sharpe vs +5.6% under IR), as the dollar’s low beta gives it a relatively higher information ratio even when its Sharpe ratio is modest. Again, this empirical pattern supports the information ratio’s separation of active return from benchmark exposure.

H. Limitations

Lagging macro indicators. The CPI and NFP releases happen at monthly intervals, with a delay of one to two weeks. The markets usually anticipate any regime changes before the actual data release, which means that the strategy will be late in taking action in terms of regimes. This was the major reason that the strategy failed in the Q4 2018 crash.

Oversimplified regime definition. The 2-by-2 regime framework considers growth and inflation as having only two states (high or low), whereas in reality, both the growth rate and inflation rate are continuously changing. Since the median is used as the classification cutoff point, there will be instances where the regime changes constantly due to their proximity to the median value.

Long-only constraint. The method can only rebalance

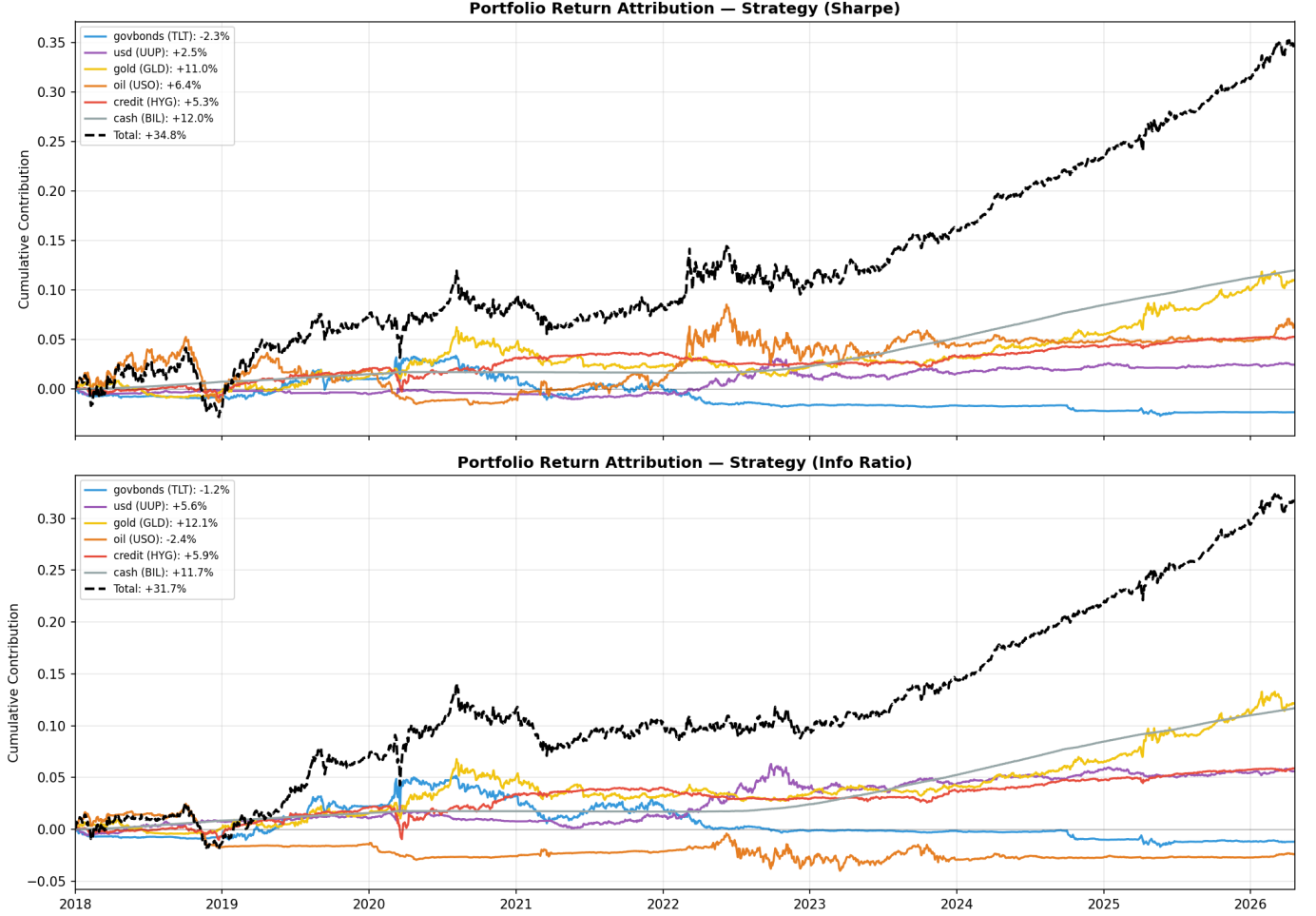


Fig. 8: Portfolio return attribution by instrument for the Sharpe strategy (top) and IR strategy (bottom). The IR strategy downweights oil (high beta) and upweights USD (low beta, higher residual information ratio), illustrating the Grinold–Kahn separation of active return from benchmark exposure.

portfolio weights between 0 and 1, as shorting is not allowed in this method. During bust regimes, the portfolio may be allocated more in defensive assets like cash and gold, but it cannot generate return by shorting risk assets. Additionally, the positivity-shifted IR weights are not identical to the unconstrained Grinold–Kahn characteristic portfolio $\mathbf{h}_A \propto \mathbf{V}^{-1}\boldsymbol{\alpha}$, which can produce short positions infeasible under our constraints; the shift-and-normalise procedure is a practical long-only approximation.

VII. CONCLUSION

By rooting regime identity in CPI and NFP, learning transition dynamics from an HMM, and deriving portfolio weights from regime-conditional risk-adjusted scores, the proposed framework eliminates hand-coded assumptions completely. The walk-forward validation over the period 2018–2026 shows that both Sharpe-based and IR-based versions deliver superior risk-adjusted performance relative to an equal-weight benchmark, with the information ratio variant achieving the highest Sharpe (1.12 vs 0.95 vs 0.78), the lowest volatility (3.4% vs 4.4% vs 8.1%), and a drawdown of -6.9% versus the benchmark’s -20.4% .

The cost is a 2.5 pp annual return shortfall concentrated entirely in the 2024–2026 credit rally that the cash-heavy defensive positioning missed — a trade-off well-suited to institutional FICC mandates where capital preservation takes precedence over return maximization.

VIII. AI DISCLAIMER

While the entire framework, model design, pipelines, and backtesting strategies are completed without the use of generative AI tools, Claude Code and Codex were used to assist with debugging the codebase, writing helper functions, and generating aesthetic images and charts. Examples of prompts used:

- (In Python) Provide a plot, with clear labels, that has time on the x-axis, with the corresponding macro regimes (defined by CPI and NFP) labelled in different colors.
- (In Python) Provide a function in `backtest.py` that learns the portfolio weights for each FICC asset at each rebalancing date, one using the Sharpe Ratio method and the other using the Information Ratio score method.

ChatGPT was also used for finding and summarizing previous research, assisting the comprehension of macroeconomic trends, and formatting the writing in LaTeX. Examples of prompts used:

- a) What are some previous prominent research done in regime identification, as well as related strategies for portfolio allocation? Give me a detailed summary with their contributions and disadvantages.
- b) Give me the equations in LaTeX used for the local linear trend model of the Kalman Filter.

In addition, portions of the manuscript were initially drafted in Mandarin Chinese and subsequently translated and revised into English.

REFERENCES

- [1] J. D. Hamilton, "A new approach to the economic analysis of nonstationary time series and the business cycle," *Econometrica*, vol. 57, no. 2, pp. 357–384, 1989.
- [2] M. Guidolin and A. Timmermann, "Asset allocation under multivariate regime switching," *J. Economic Dynamics and Control*, vol. 31, no. 11, pp. 3503–3544, 2007.
- [3] A. Ang and A. Timmermann, "Regime changes and financial markets," *Annual Review of Financial Economics*, vol. 4, pp. 313–337, 2012.
- [4] A. Ilmanen, *Expected Returns*. Wiley, 2011.
- [5] R. C. Grinold and R. N. Kahn, *Active Portfolio Management: A Quantitative Approach for Producing Superior Returns and Controlling Risk*, 2nd ed. New York: McGraw-Hill, 1999.